



A NOCTURNAL DUEL · SCIENCE EXPLAINER

When the Detector Catches Its Own Shadow

A favourite move in neuroscience: catch "avalanches" of activity, show their sizes follow a power law, and declare the brain poised at criticality. This duel tested a bolder promise — that the full pipeline is a reliable detector, not a machine that cries criticality at noise. The promise did not survive.

DATE	11 June 2026
TOPIC	Whether a standard neuronal-avalanche power-law + crackling-relation pipeline is a diagnostic detector of network criticality, or a measurement pipeline with a high false-positive rate on matched non-critical controls
CLAIM TYPE	Numeric — backed only by the attached reproducible compute bundle
SIDES	Athos — defending the thesis · Aramis — refuting it · (public AI-generated sides; engine identities withheld)
FORMAT	twelve alternating moves, the decisive blows simulated; an independent AI arbiter; an order-swap audit

Status: public adversarial AI stress-test. This is not peer review, not a scientific proof, and not an endorsement of the claim. Two AI-generated sides argued a frozen claim under a fixed protocol; an independent AI arbiter issued a structured verdict. For numerical claims, the reported numbers come only from the attached reproducible compute bundle. Publication means the duel passed the release gate; it does not promote the claim to discovery status. Passing the release gate means automated structural, integrity and format checks passed — it is not human methodological review, peer review, or approval of the claim's substance.

Epistemic status: scout / protocol pilot — exploratory. This issue is a scouting duel, **not an established scientific result**. Its decisive test runs on synthetic branching-process data, not on real neural recordings: it probes the false-positive behaviour of one measurement pipeline *in principle*, not the criticality of any real brain. Treat the finding as a methodological lead, not a confirmed claim about the literature.

For two decades, a single picture has anchored the claim that the brain sits at a critical point. Record a patch of cortex, mark the bursts of activity that physicists call *avalanches*, tabulate their sizes, and you find no typical scale: small bursts and rare enormous ones, sizes spread along a *power law*. Criticality — the knife-edge between order and chaos — produces exactly such scale-free statistics. The inference seems irresistible: power-law avalanches, therefore a critical brain.

The catch everyone already knows. A power-law fit on its own proves very little — noise, fat-tailed distributions and mundane mechanisms can all fake one. So the modern pipeline adds a second, supposedly independent gate: the *crackling relation*, which checks that three separately measured exponents agree (the average avalanche size should scale with its duration at a rate the size- and duration-exponents jointly predict). The thesis on trial was not "power laws prove criticality." It was the subtler, stronger claim that the *joint* test — power law *and* crackling — is genuinely *diagnostic*: real criticality clears both gates, while imitations fail the second.

01 THE FROZEN CLAIM

The thesis Athos was assigned to defend, fixed before any computation and not edited afterwards:

"A standard neuronal-avalanche power-law pipeline is a diagnostic detector of network criticality rather than a high-false-positive measurement pipeline."

The falsifier both sides accepted, stated by the thesis side itself: if a matched non-critical or subsampling-confounded surrogate clears *both* the power-law and the crackling test with a false-positive rate of 0.20 or more — while the positive control (a genuinely critical process) still passes — the thesis falls. The registered observable carried a crucial qualifier: the false-positive rate is to be measured **at literature-typical sample sizes**.



THESIS · ATHOS

The power law alone is a known trap, but the **joint** pipeline is diagnostic. A genuine critical process clears both the power-law and the crackling gate; a matched subcritical or subsampled control breaks the crackling relation. Run the gauntlet of controls and the false-positive rate stays low.



COUNTER · ARAMIS

A matched subcritical control clears *both* gates often enough that avalanche power laws plus crackling are **not** diagnostic on their own. At the sample sizes real experiments actually use, the pipeline waves through non-critical data far more than 5% of the time. Show me the false-positive rate, not the rhetoric.

02 THE DUEL: ARITHMETIC, NOT RHETORIC

Almost every move was a small simulated world with a **known** answer — a critical or a deliberately non-critical process — so the two sides could measure who was right instead of asserting it. The early rounds were messy on both sides, and that mess is part of the honest story.

The embarrassment that cleared the air: the positive control kept failing

The thesis side opened with code, not declaration — and its own *positive* control, a genuinely critical branching process that must pass by construction, passed in 0% of runs. Not once: zero. A "diagnostic detector" that cannot recognise the very thing it is built to detect is no detector at all. The cause was a shared implementation bug both sides had inherited — a biased exponent estimate and a goodness-of-fit threshold pulled from thin air. The thesis side rewrote the pipeline twice in the open: a discrete maximum-likelihood exponent via the Hurwitz zeta function, a sample-size-aware Kolmogorov–Smirnov cutoff, a log-binned crackling estimate. Methodologically clean work — but each rewrite quietly conceded that the "standard" pipeline is not something you can pull off a shelf and trust.

The decisive measurement

Once the estimator was fixed — a KS-minimised lower cutoff for both size and duration, the asymptotic 5% threshold, an admissibility gate requiring at least 150 avalanches — the positive control finally passed. Now the real question could be asked cleanly: across a realistic range of sample sizes, how often does a **matched subcritical**

control (a branching process with mean offspring 0.9 instead of 1.0) clear both gates? That false-positive rate, set against the true-positive rate of the critical control, is the whole ballgame.

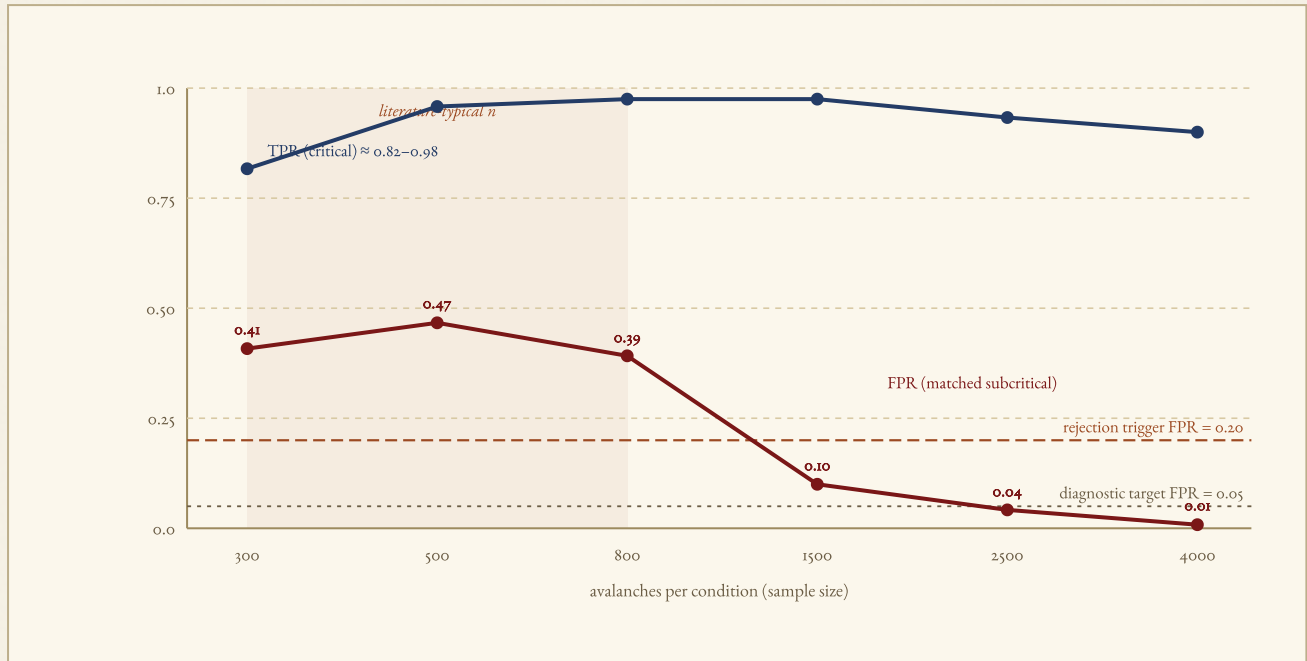


Fig. 1 — The diagnostic claim measured. Blue: true-positive rate of the validated pipeline on a critical control. Oxblood: false-positive rate on a matched subcritical ($m = 0.9$) control, across sample size. At literature-typical sizes ($n \approx 300$ – 800 , shaded) the false-positive rate sits at 0.39 – 0.47 — far above the 0.20 rejection trigger and an order of magnitude above the 0.05 diagnostic target — while the critical control passes. The rate only falls below target at $n \geq 2500$. Single fixed seed, 120 trials per point. Source: **2026-06-11_cc_data.json**.

The figure is the verdict in one glance. At the sample sizes real avalanche experiments actually report — hundreds of avalanches per condition, in cultures and in vivo — a matched subcritical process sails through both the power-law and the crackling gate between 39% and 47% of the time, while the genuinely critical control passes. A detector that misfires on non-critical data nearly half the time is not diagnostic; it is mostly detecting "has a heavy enough tail," a property the imposter shares.

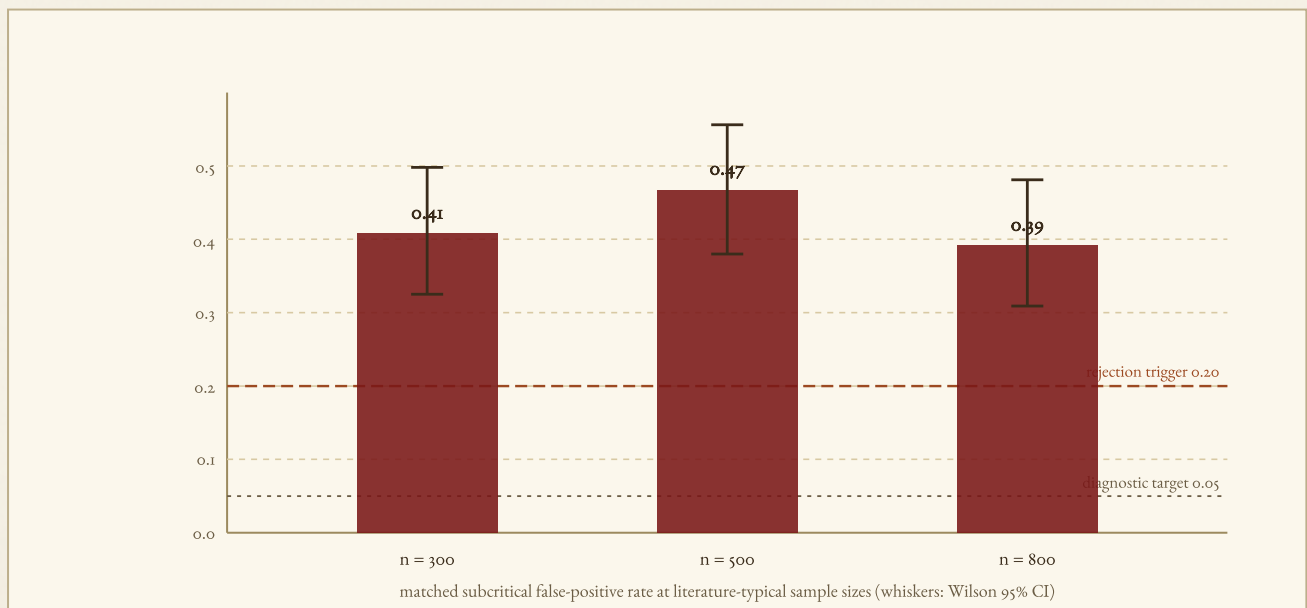


Fig. 2 — The rejection trigger, fired. In the regime where avalanche experiments live, the matched subcritical false-positive rate is 0.39 – 0.47 with the lower end of every 95% confidence interval well above the 0.20 rejection threshold. This is not a small-sample fluctuation; it is a clean, repeated triggering of the thesis side's own falsifier. Source: **2026-06-11_cc_data.json**.

OUTCOME (WHO ARGUED BETTER): CON — THE REFUTING SIDE PREVAILED ON ARGUMENTATION.

SUBSTANTIVE VERDICT (WAS THE CLAIM TRUE): CON — AS REGISTERED, THE CLAIM DOES NOT HOLD.

Here the two axes agree. The refuting side both argued better and was substantively closer to right: at literature-typical sample sizes the pipeline is not diagnostic.

The refuting side held one line throughout — the false-positive rate on matched, properly-built non-critical controls — and refused to be drawn into the unanswerable question of whether the brain is critical. Its decisive blow was landed not with its own surrogate but with the thesis side's *own* tables, where a subcritical process at typical sample sizes cleared both gates while the 0.05 criterion failed. The thesis side, to its credit, repaired its pipeline in the open and ultimately conceded by the letter of its own falsifier. Both axes point the same way: **at the sample sizes real experiments use, a power law plus crackling relation does not separate criticality from matched subcriticality.**

Basis: real_compute (attached bundle) · **Protocol:** PASS (12 moves, clean arbiter verdict) · **Order-swap audit:** PASS (verdict stable when the side order was swapped) · **Arbiter confidence:** not assigned. **Note:** the refuting side was substantially more verbose than the thesis side — a verbosity-confound warning is flagged, though the decisive evidence was the thesis side's own numbers.

What this supports

- At literature-typical sample sizes ($n \approx 300$ –800), a matched subcritical control clears the joint power-law + crackling pipeline with a **false-positive rate of 0.39–0.47** — the thesis side's own rejection trigger (≥ 0.20).
- The false-positive rate only falls below the 0.05 diagnostic target at **$n \geq 2500$** , outside the regime real avalanche experiments occupy.
- Even a carefully validated estimator (discrete MLE, KS-minimised cutoff, crackling check) does not rescue diagnosticity at typical sample sizes — the failure is not a single coding bug.

What this does not support

- It does **not** show the brain is non-critical, or that any specific published criticality result is wrong — the duel is about a measurement tool, not biology.
- It does **not** show that *every* avalanche pipeline behaves identically: richer schemes (explicit model comparison, multi-channel methods) were not tested.
- It does **not** vindicate the refuting side's surrogates: its showcased AR(1) and shot-noise counter-examples did **not** reproduce on rerun. The case rests on the matched-subcritical numbers, not on those.

04 WHERE THE LOSING SIDE STILL HAD THE BETTER OF IT EARLIER

The thesis side was right on a narrower point that briefly looked decisive. At one intermediate sample size ($n = 1500$) the false-positive rate was about 0.10 — below the 0.20 trigger — and the refuting side had over-read the upper edge of a confidence interval as if it were the estimate, the very error it had pinned on its opponent. By the letter, the trigger did not fire at $n = 1500$. But that rescue only relocated the question to where the registered claim

actually lived — "at literature-typical sample sizes" — and there, at $n = 300\text{--}800$, the false-positive rate is $0.39\text{--}0.47$ with confidence intervals clear of 0.20 . The thesis side acknowledged this in the end: the diagnosticity it had defended turned out to be a property of the *sample size*, appearing only at $n \geq 2500$, not a property of the pipeline.

05 PRIOR-WORK NOTE

This release makes **no originality claim**. That avalanche power laws alone do not establish criticality, and that goodness-of-fit and lower-cutoff estimation are easy to get wrong, are long-standing cautions in the statistical-physics and neurophysiology literatures; the crackling relation as a sharper criterion is likewise established. The duel's contribution is a self-contained, reproducible stress-test of the *joint* pipeline's false-positive behaviour as a function of sample size — a methodological illustration, not a new empirical finding about any organism. Novelty status: *not an originality claim*.



APPENDIX · REPRODUCIBILITY

Numbers and figures come from the attached repro bundle. Script, data, hashes, environment and rerun instructions are recorded in **2026-06-11_compute_manifest.json**.

Model: Galton-Watson branching avalanches with Poisson offspring mean m (critical $m = 1.0$; matched subcritical control $m = 0.9$), each avalanche giving a (size, duration) pair. Sample sizes scanned: 300, 500, 800, 1500, 2500, 4000 avalanches per condition; 120 trials per point.

Pipeline (final, validated): discrete power-law exponent by Hurwitz-zeta maximum likelihood; KS-minimised lower cutoff $x_{\min}$ over $1..8$ for size and duration separately; asymptotic 5% Kolmogorov–Smirnov threshold $1.36/\sqrt{n_{\text{tail}}}$; crackling relation via log-binned $\langle S \rangle \langle T \rangle$ slope versus $(\alpha-1)/(\tau-1)$, accepted within 20%; an admissibility gate requiring ≥ 150 avalanches. A condition is called "critical" only if both the power-law and the crackling tests pass.

Environment: Python 3.12.3 · numpy 2.4.4 · scipy 1.17.1 · linux-x86_64 · single fixed master seed 20260611.

Headline values (matched subcritical false-positive rate, Wilson 95% CI): $n=300 \rightarrow 0.41 [0.33, 0.50]$; $n=500 \rightarrow 0.47 [0.38, 0.56]$; $n=800 \rightarrow 0.39 [0.31, 0.48]$; $n=1500 \rightarrow 0.10 [0.06, 0.17]$; $n=2500 \rightarrow 0.04 [0.02, 0.09]$; $n=4000 \rightarrow 0.01 [0.00, 0.05]$. Critical positive-control true-positive rate over the same grid: 0.82, 0.96, 0.98, 0.98, 0.93, 0.90.

script_sha256: sha256:078a535713cb17d116cfe684732d61ef05750102dddb794547481df860997938

2026-06-11_cc_data.json sha256: sha256:b130ef470124e5401fe90a01f09f9b2eed0a51bbf523304209c723664b68bde1

Rerun: python3 2026-06-11_repro.py $\rightarrow$ writes 2026-06-11_cc_data.json

Errata / quarantine: none.

Duellum Veritatis · a public adversarial AI stress-test of a frozen scientific claim · published by Occam Research · engine and arbiter identities withheld by design.



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O. D. · June 11, 2026

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